

Headline metrics — std log-IV space (overall, n\_test=1007) (pred\_len=21)

| display               | n_seeds | RMSE<br>( $\pm \sigma_{\text{seed}}$ )<br>[95% CI] | MSE    | MAE    | R <sup>2</sup> vs pers<br>[95% CI] | var ratio | params  | val_loss |
|-----------------------|---------|----------------------------------------------------|--------|--------|------------------------------------|-----------|---------|----------|
| dlinear               | 1       | 0.4696<br>[0.3211, 0.6630]                         | 0.2205 | 0.2910 | +0.0432<br>[-0.0389, +0.1779]      | 0.849     | 403,200 | 0.1625   |
| persistence           | —       | 0.4801<br>[0.3359, 0.6659]                         | 0.2305 | 0.3001 | — (reference)                      | 0.996     | —       | —        |
| tucker_dlinear        | 1       | 0.4822<br>[0.3351, 0.6720]                         | 0.2325 | 0.3054 | -0.0085<br>[-0.1238, +0.1337]      | 0.768     | 8,880   | 0.1579   |
| patchtst              | 1       | 0.4830<br>[0.3185, 0.6952]                         | 0.2333 | 0.2940 | -0.0119<br>[-0.1349, +0.1726]      | 0.859     | 5,894   | 0.1583   |
| var                   | 1       | 0.5043<br>[0.3265, 0.7301]                         | 0.2544 | 0.2976 | -0.1035<br>[-0.2450, +0.1545]      | 0.828     | —       | —        |
| gwn                   | 1       | 0.5056<br>[0.3386, 0.7216]                         | 0.2556 | 0.3141 | -0.1088<br>[-0.2711, +0.0976]      | 0.805     | 11,369  | 0.1549   |
| hot/kronecker_sum     | 1       | 0.5170<br>[0.3548, 0.7273]                         | 0.2673 | 0.3292 | -0.1597<br>[-0.3220, -0.0068]      | 0.887     | 84,949  | 0.1751   |
| hot/kronecker_product | 1       | 0.5306<br>[0.3465, 0.7585]                         | 0.2816 | 0.3287 | -0.2215<br>[-0.4529, +0.0315]      | 0.705     | 167,765 | 0.1708   |

Headline metrics — COVID (n\_test=254) (pred\_len=21)

| display               | n_seeds | RMSE<br>( $\pm \sigma_{\text{seed}}$ )<br>[95% CI] | MSE    | MAE    | R <sup>2</sup> vs pers<br>[95% CI] | var ratio | params  | val_loss |
|-----------------------|---------|----------------------------------------------------|--------|--------|------------------------------------|-----------|---------|----------|
| dlinear               | 1       | 0.7340<br>[0.3339, 1.1584]                         | 0.5387 | 0.4064 | +0.0621<br>[-0.0390, +0.3847]      | 0.891     | 403,200 | 0.1625   |
| tucker_dlinear        | 1       | 0.7529<br>[0.3752, 1.1732]                         | 0.5669 | 0.4401 | +0.0131<br>[-0.1202, +0.2636]      | 0.802     | 8,880   | 0.1579   |
| persistence           | —       | 0.7579<br>[0.3898, 1.1661]                         | 0.5744 | 0.4455 | — (reference)                      | 1.043     | —       | —        |
| patchtst              | 1       | 0.7808<br>[0.3514, 1.2437]                         | 0.6097 | 0.4312 | -0.0615<br>[-0.2181, +0.2987]      | 0.900     | 5,894   | 0.1583   |
| gwn                   | 1       | 0.8129<br>[0.3756, 1.2820]                         | 0.6608 | 0.4629 | -0.1504<br>[-0.3843, +0.2360]      | 0.829     | 11,369  | 0.1549   |
| hot/kronecker_sum     | 1       | 0.8149<br>[0.3732, 1.2718]                         | 0.6641 | 0.4731 | -0.1562<br>[-0.4074, +0.1755]      | 0.863     | 84,949  | 0.1751   |
| var                   | 1       | 0.8342<br>[0.3843, 1.3063]                         | 0.6959 | 0.4640 | -0.2115<br>[-0.4223, +0.2001]      | 0.883     | —       | —        |
| hot/kronecker_product | 1       | 0.8665<br>[0.4057, 1.3450]                         | 0.7508 | 0.5080 | -0.3071<br>[-0.7055, +0.0782]      | 0.704     | 167,765 | 0.1708   |

Headline metrics — Reflation calm (n\_test=252) (pred\_len=21)

| display               | n_seeds | RMSE<br>( $\pm \sigma_{\text{seed}}$ )<br>[95% CI] | MSE    | MAE    | R <sup>2</sup> vs pers<br>[95% CI] | var ratio | params  | val_loss |
|-----------------------|---------|----------------------------------------------------|--------|--------|------------------------------------|-----------|---------|----------|
| var                   | 1       | 0.2739<br>[0.2278, 0.3066]                         | 0.0750 | 0.2025 | +0.2194<br>[+0.1090, +0.3619]      | 0.805     | —       | —        |
| tucker_dlinear        | 1       | 0.2808<br>[0.2454, 0.3088]                         | 0.0788 | 0.2113 | +0.1799<br>[+0.0123, +0.2901]      | 0.728     | 8,880   | 0.1579   |
| patchtst              | 1       | 0.2826<br>[0.2369, 0.3173]                         | 0.0799 | 0.2083 | +0.1692<br>[+0.0923, +0.2330]      | 0.879     | 5,894   | 0.1583   |
| dlinear               | 1       | 0.2948<br>[0.2459, 0.3299]                         | 0.0869 | 0.2180 | +0.0958<br>[-0.0420, +0.2269]      | 0.840     | 403,200 | 0.1625   |
| hot/kronecker_product | 1       | 0.3034<br>[0.2500, 0.3435]                         | 0.0920 | 0.2270 | +0.0427<br>[-0.1527, +0.2348]      | 0.715     | 167,765 | 0.1708   |
| gwn                   | 1       | 0.3049<br>[0.2499, 0.3431]                         | 0.0929 | 0.2265 | +0.0332<br>[-0.0857, +0.1826]      | 0.868     | 11,369  | 0.1549   |
| persistence           | —       | 0.3100<br>[0.2611, 0.3477]                         | 0.0961 | 0.2246 | — (reference)                      | 0.997     | —       | —        |
| hot/kronecker_sum     | 1       | 0.3238<br>[0.2722, 0.3632]                         | 0.1049 | 0.2417 | -0.0909<br>[-0.1836, -0.0060]      | 0.882     | 84,949  | 0.1751   |

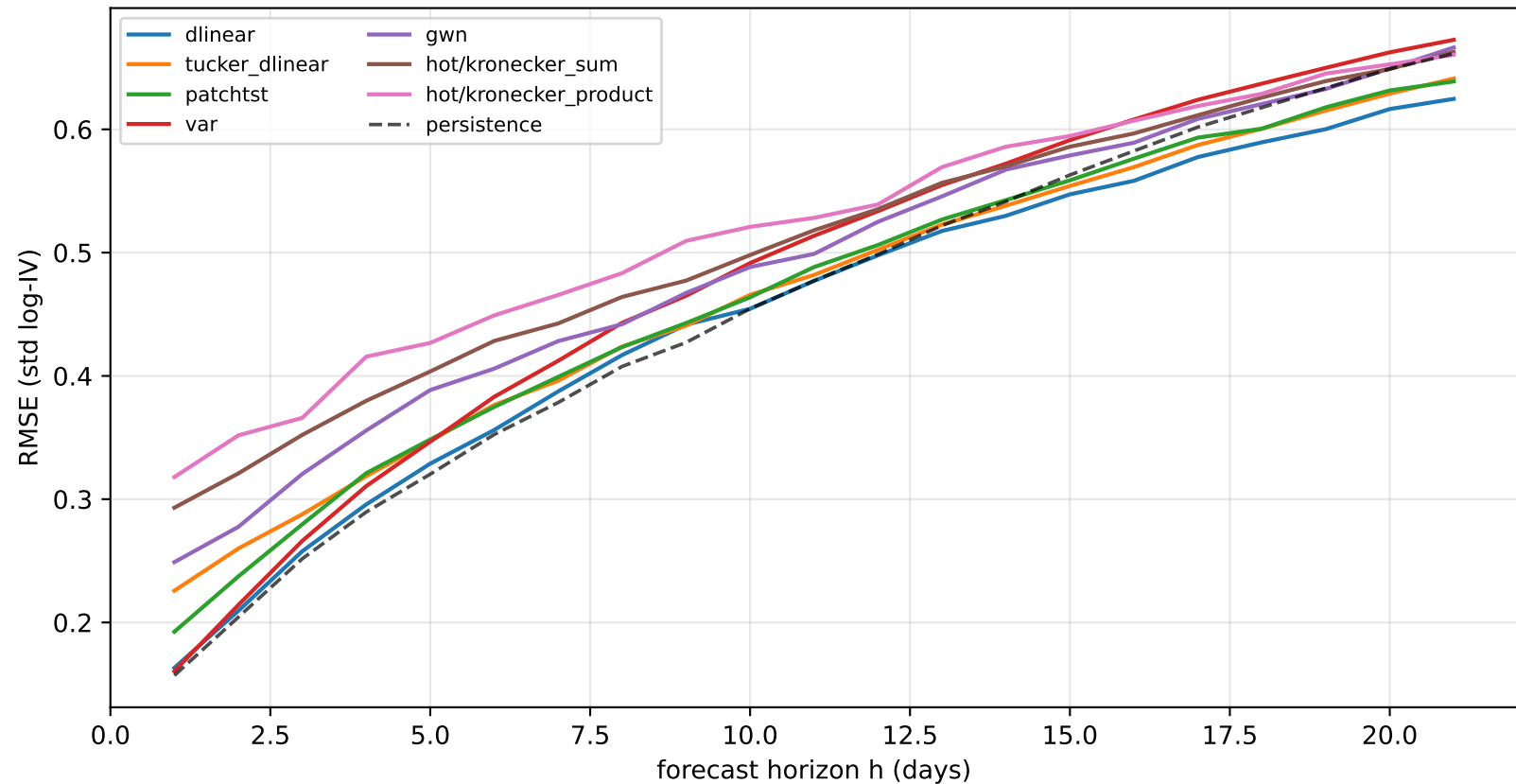
Headline metrics — Bear 2022 (n\_test=251) (pred\_len=21)

| display               | n_seeds | RMSE<br>( $\pm \sigma_{\text{seed}}$ )<br>[95% CI] | MSE    | MAE    | R <sup>2</sup> vs pers<br>[95% CI] | var ratio | params  | val_loss |
|-----------------------|---------|----------------------------------------------------|--------|--------|------------------------------------|-----------|---------|----------|
| patchtst              | 1       | 0.3829<br>[0.3197, 0.4558]                         | 0.1466 | 0.2903 | +0.0284<br>[-0.1602, +0.1514]      | 0.932     | 5,894   | 0.1583   |
| persistence           | —       | 0.3884<br>[0.3254, 0.4580]                         | 0.1509 | 0.2936 | — (reference)                      | 1.038     | —       | —        |
| var                   | 1       | 0.3923<br>[0.3194, 0.4751]                         | 0.1539 | 0.2921 | -0.0199<br>[-0.3387, +0.2123]      | 0.865     | —       | —        |
| dlinear               | 1       | 0.4027<br>[0.3356, 0.4764]                         | 0.1622 | 0.3042 | -0.0748<br>[-0.3649, +0.1275]      | 0.892     | 403,200 | 0.1625   |
| gwn                   | 1       | 0.4217<br>[0.3541, 0.4981]                         | 0.1778 | 0.3274 | -0.1785<br>[-0.4907, +0.0497]      | 0.965     | 11,369  | 0.1549   |
| hot/kronecker_sum     | 1       | 0.4271<br>[0.3602, 0.4995]                         | 0.1824 | 0.3355 | -0.2088<br>[-0.4405, +0.0045]      | 1.104     | 84,949  | 0.1751   |
| hot/kronecker_product | 1       | 0.4356<br>[0.3571, 0.5235]                         | 0.1897 | 0.3339 | -0.2576<br>[-0.7106, +0.0577]      | 0.794     | 167,765 | 0.1708   |
| tucker_dlinear        | 1       | 0.4370<br>[0.3612, 0.5225]                         | 0.1910 | 0.3303 | -0.2658<br>[-0.6606, +0.0384]      | 0.835     | 8,880   | 0.1579   |

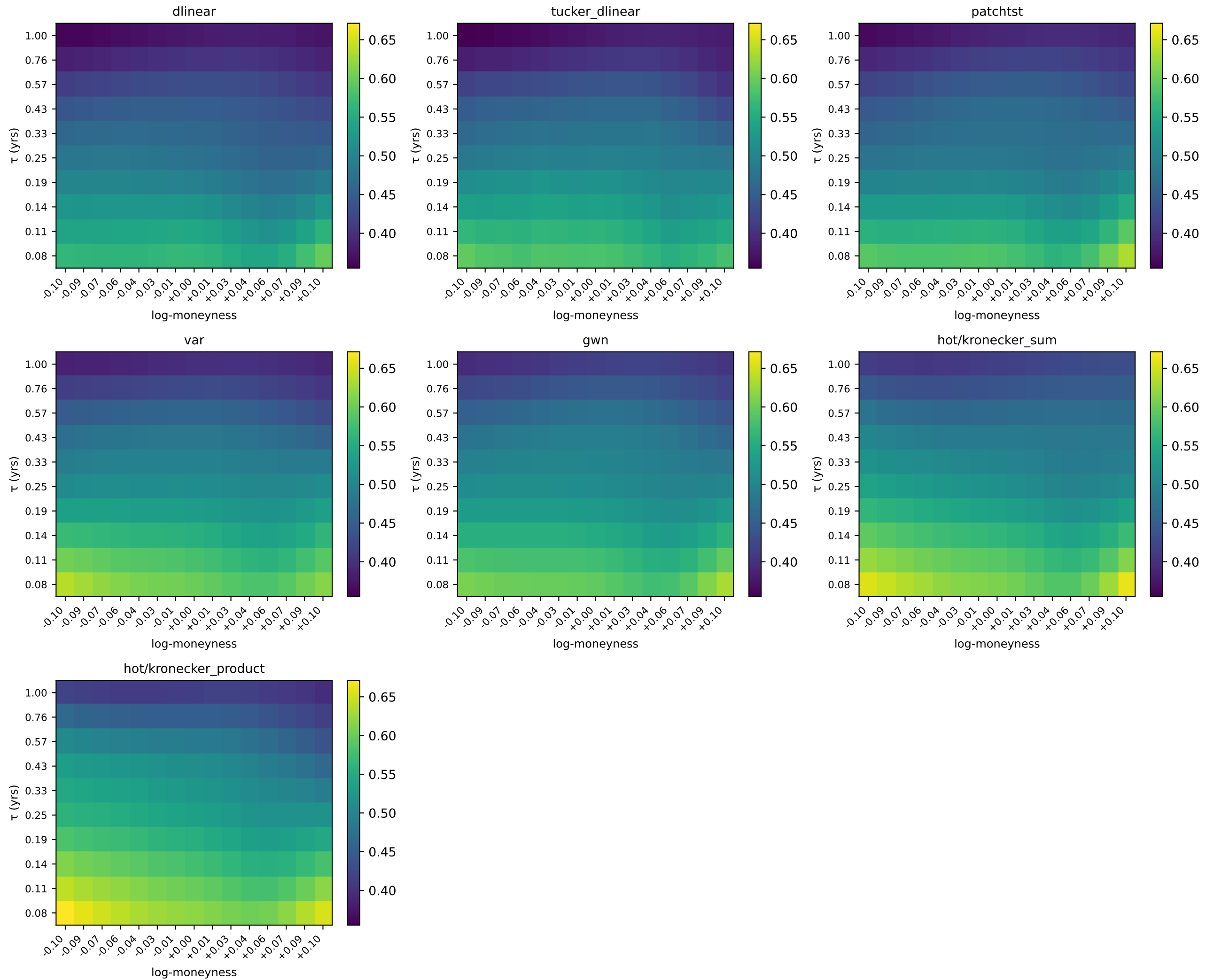
Headline metrics — Normalisation (n\_test=250) (pred\_len=21)

| display               | n_seeds | RMSE<br>( $\pm \sigma_{\text{seed}}$ )<br>[95% CI] | MSE    | MAE    | R <sup>2</sup> vs pers<br>[95% CI] | var ratio | params  | val_loss |
|-----------------------|---------|----------------------------------------------------|--------|--------|------------------------------------|-----------|---------|----------|
| gwn                   | 1       | 0.2931<br>[0.2553, 0.3320]                         | 0.0859 | 0.2378 | +0.1101<br>[-0.0633, +0.2979]      | 0.841     | 11,369  | 0.1549   |
| var                   | 1       | 0.2957<br>[0.2627, 0.3424]                         | 0.0874 | 0.2300 | +0.0943<br>[-0.0989, +0.2337]      | 0.935     | —       | —        |
| hot/kronecker_product | 1       | 0.2968<br>[0.2603, 0.3383]                         | 0.0881 | 0.2440 | +0.0878<br>[-0.1438, +0.2787]      | 0.856     | 167,765 | 0.1708   |
| tucker_dlinear        | 1       | 0.2988<br>[0.2557, 0.3507]                         | 0.0893 | 0.2386 | +0.0751<br>[-0.1317, +0.2754]      | 0.792     | 8,880   | 0.1579   |
| dlinear               | 1       | 0.3010<br>[0.2556, 0.3618]                         | 0.0906 | 0.2339 | +0.0616<br>[-0.0642, +0.2142]      | 0.889     | 403,200 | 0.1625   |
| patchtst              | 1       | 0.3040<br>[0.2658, 0.3498]                         | 0.0924 | 0.2449 | +0.0426<br>[-0.1544, +0.2084]      | 0.910     | 5,894   | 0.1583   |
| persistence           | —       | 0.3107<br>[0.2681, 0.3708]                         | 0.0965 | 0.2351 | — (reference)                      | 1.034     | —       | —        |
| hot/kronecker_sum     | 1       | 0.3366<br>[0.2908, 0.3967]                         | 0.1133 | 0.2647 | -0.1734<br>[-0.5127, +0.0837]      | 1.031     | 84,949  | 0.1751   |

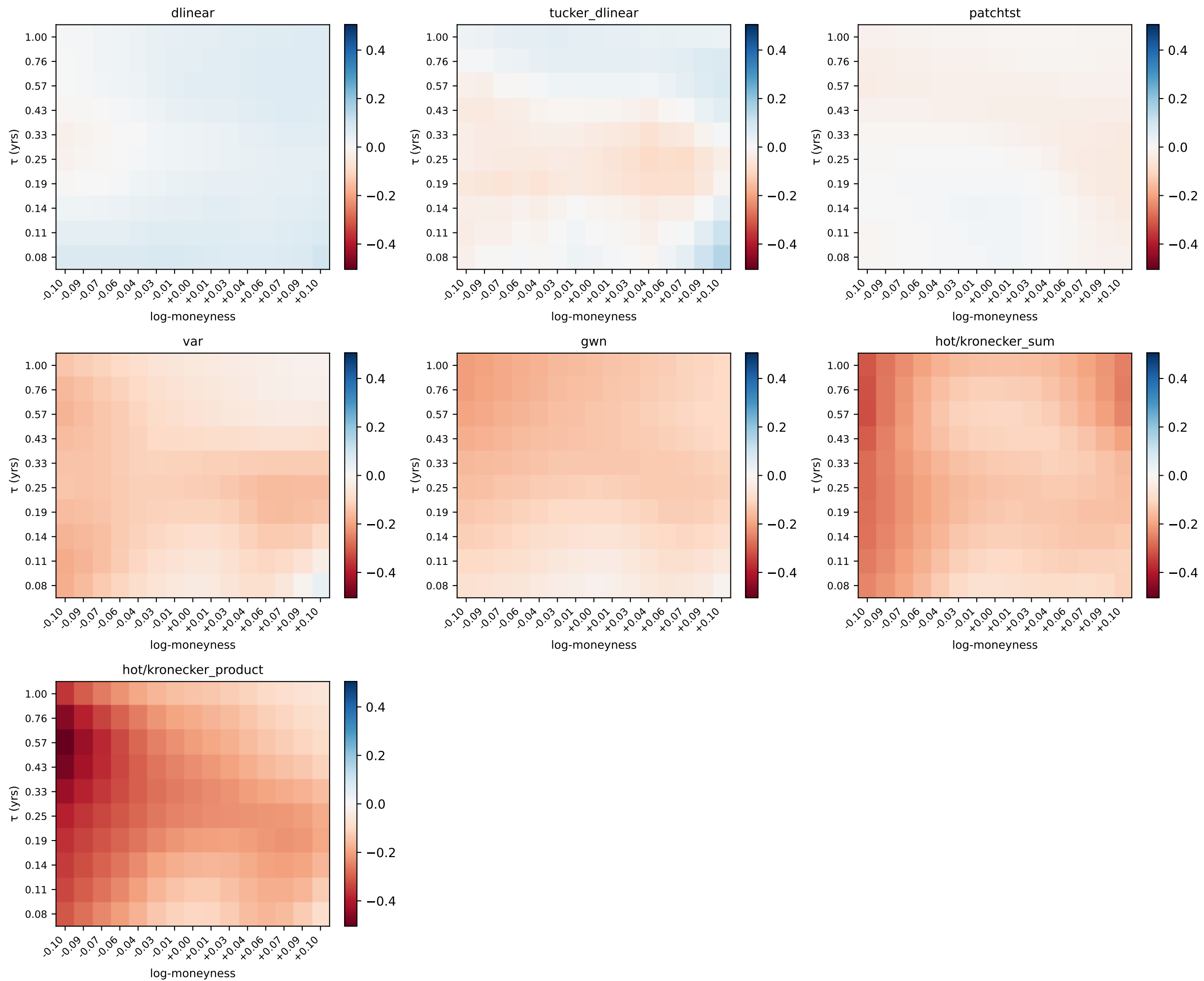
# RMSE vs forecast horizon



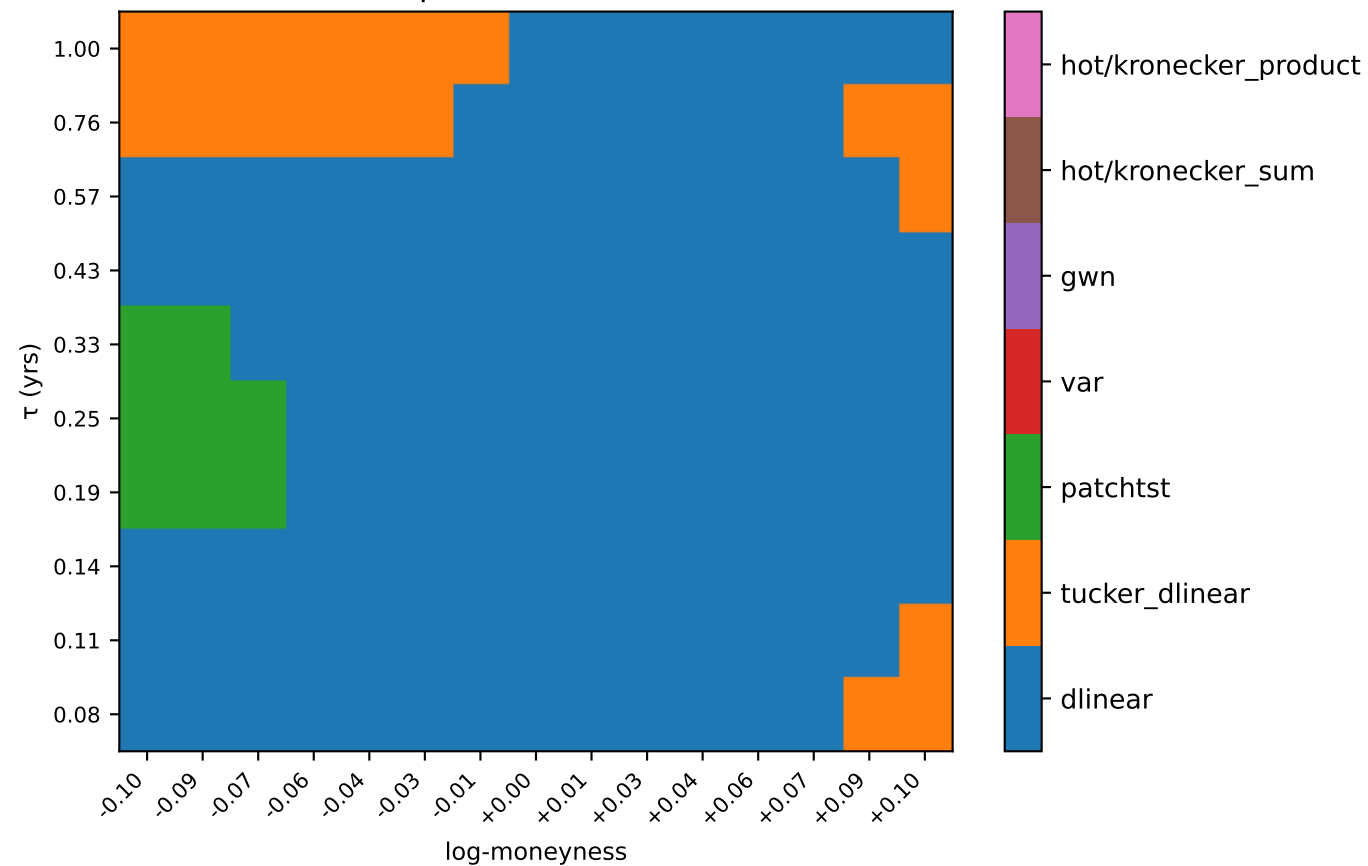
Per-cell RMSE (std log-IV) — each panel is one model



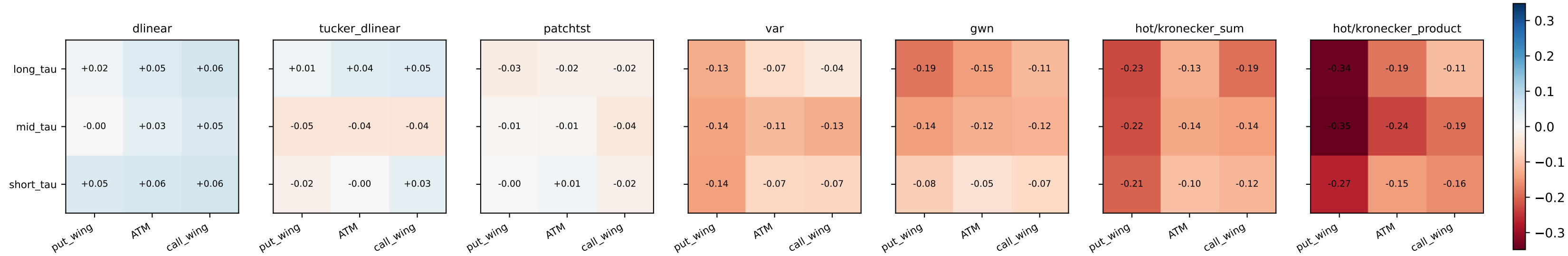
$R^2$  vs persistence per cell (red = worse than persistence)



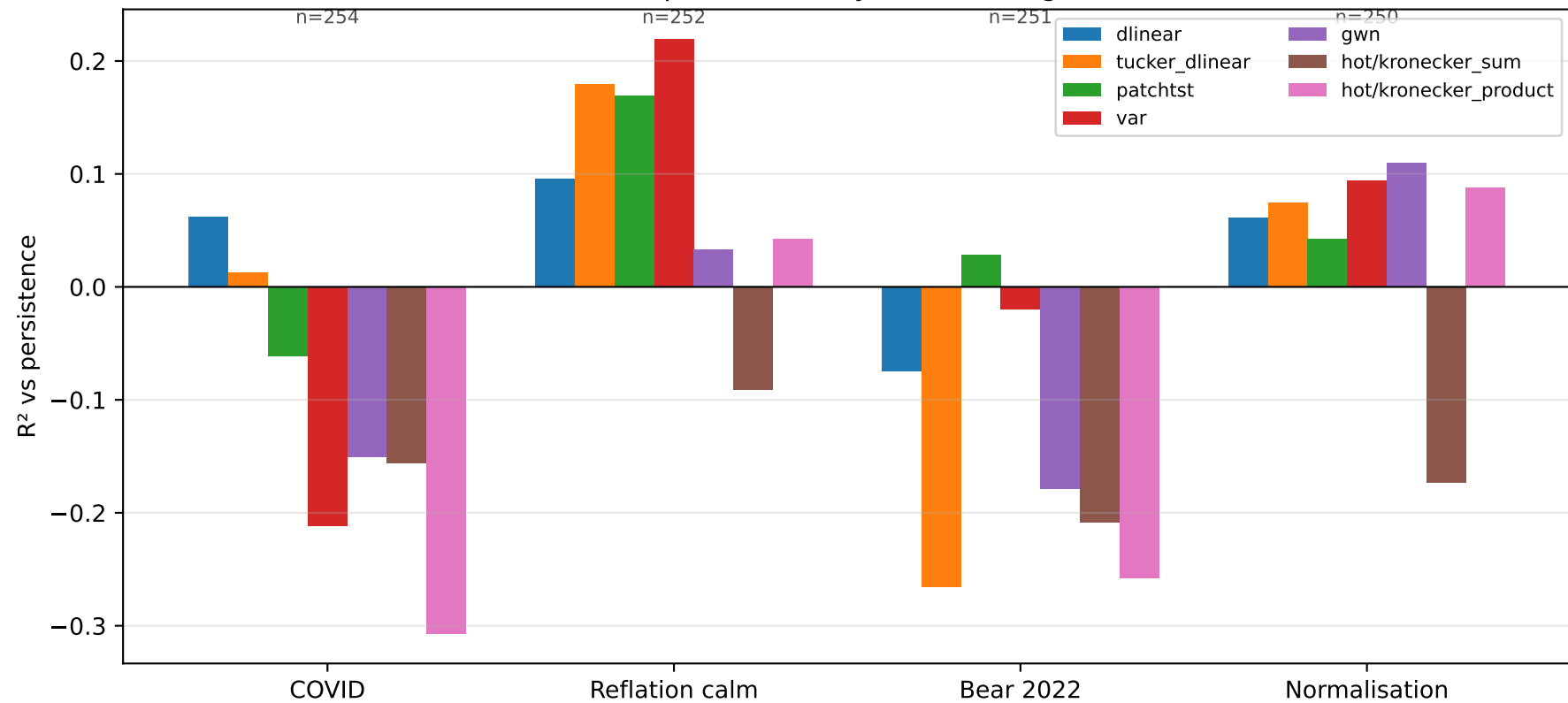
# Winner per cell (lowest RMSE)



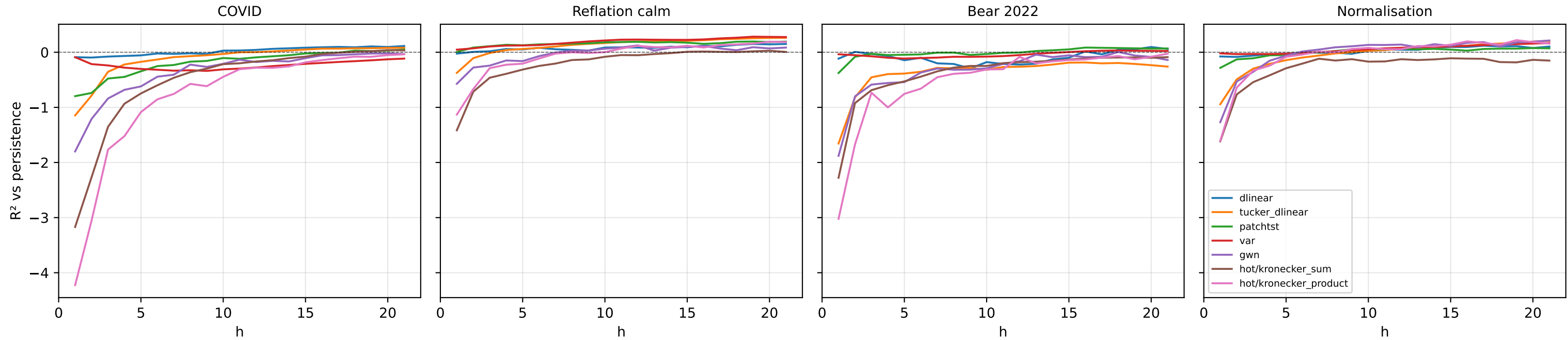
R<sup>2</sup> vs persistence by surface zone

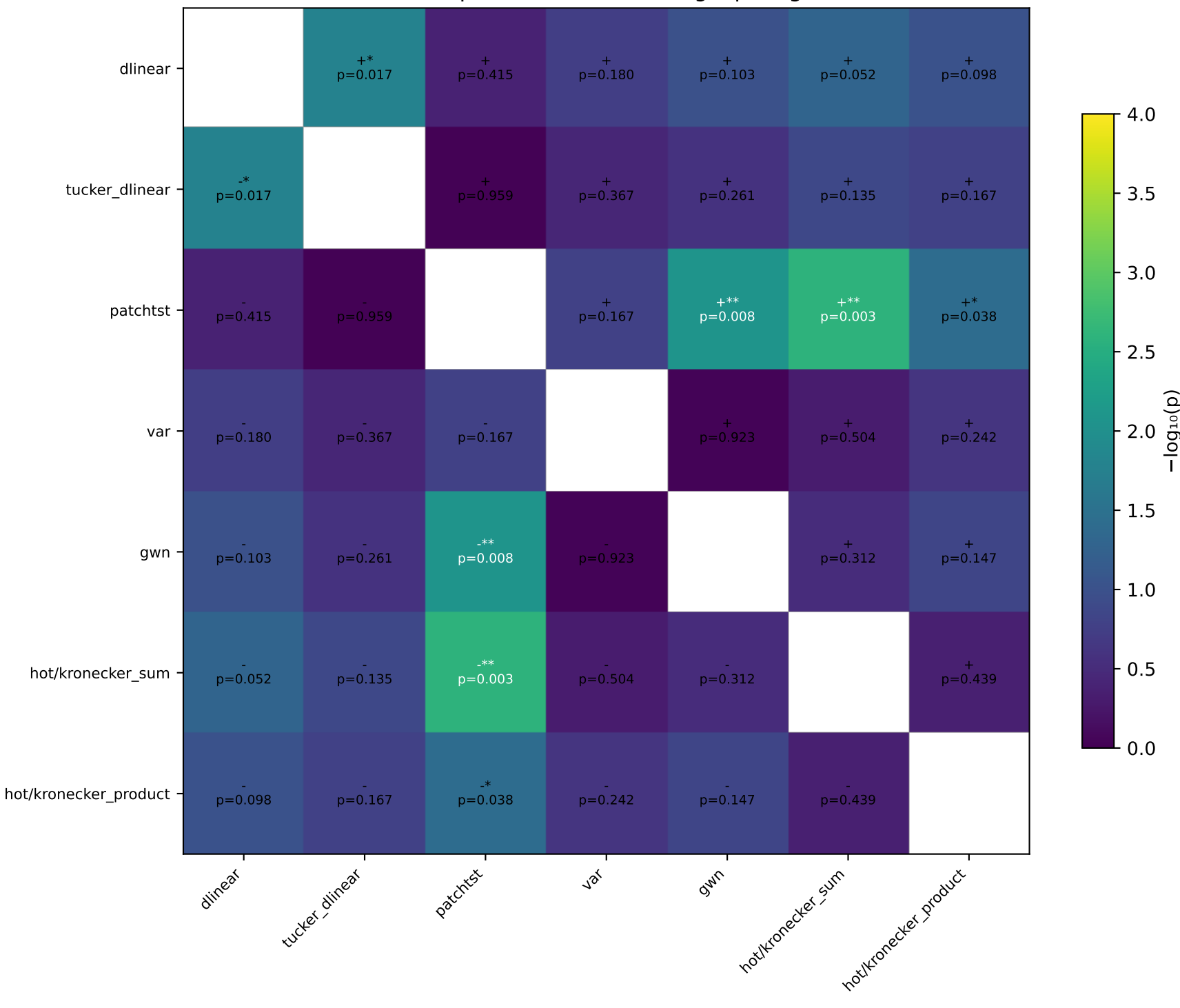


# R<sup>2</sup> vs persistence by calendar regime

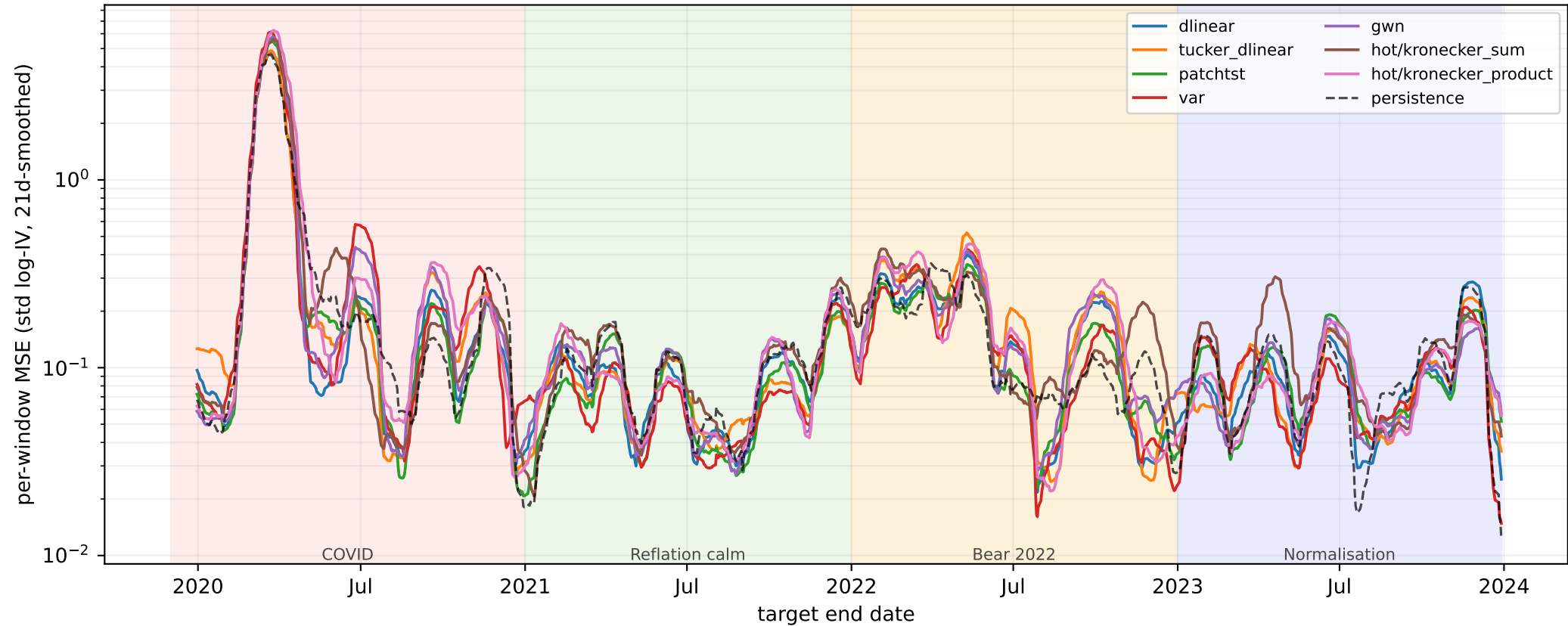


$R^2$  vs persistence by horizon, faceted by regime

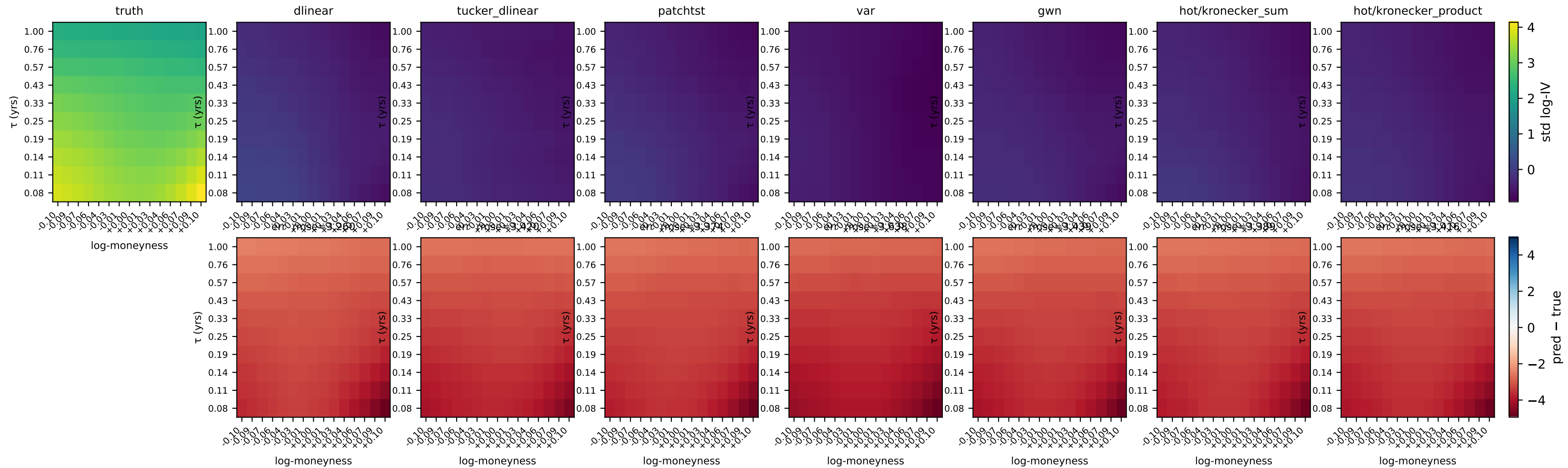


Diebold-Mariano pairwise — color:  $-\log_{10}(p)$ , sign: row vs col

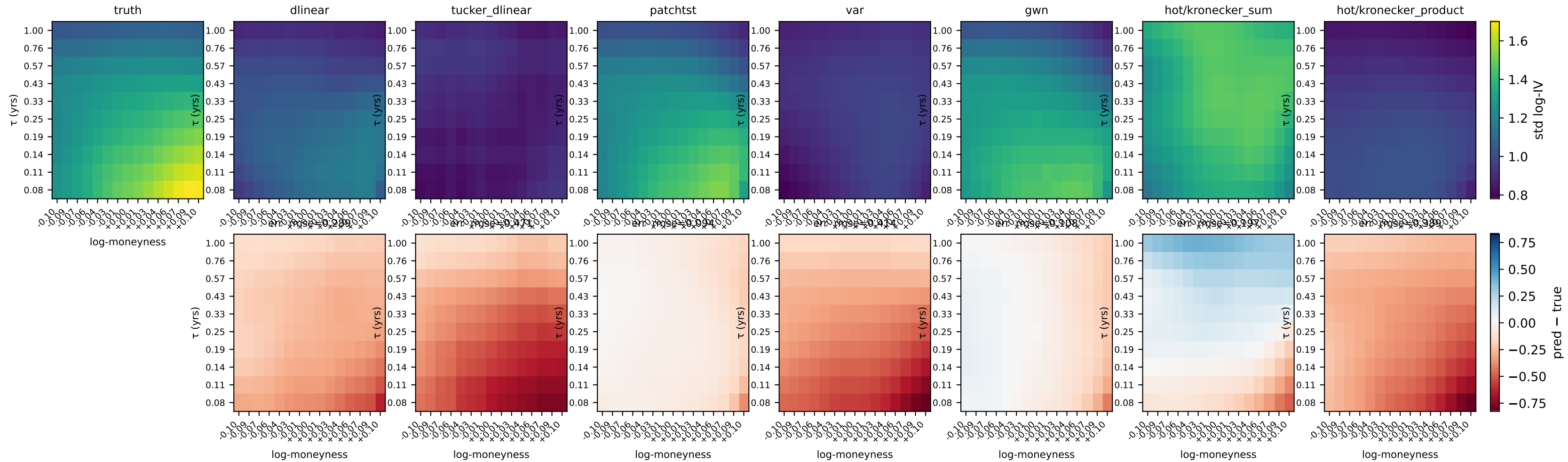
Per-window MSE over time, regime-shaded



Example forecast — covid: target 2020-03-23 (h=21)



Example forecast — bear: target 2022-06-15 (h=21)



Example forecast — calm: target 2023-08-15 (h=21)

